



Programme: Data Science and Business Analytics

CALCULUS

Lecture 4 (28.10.2026)

The number e . Cantor's theorem. Bolzano-Weierstrass theorem.

1. Number e .

Let us consider the sequence

$$x_n = \left(1 + \frac{1}{n}\right)^n, \quad n \in \mathbb{N}. \quad (1)$$

We can calculate that

$$x_1 = 2, \quad x_2 = 2.25, \quad x_3 = 2.37, \quad x_4 = 2.44, \dots$$

Theorem 1. *The sequence (1) is convergent.*

Proof. To prove that x_n is convergent let us consider the sequence

$$y_n = x_n \cdot \left(1 + \frac{1}{n}\right) = \left(1 + \frac{1}{n}\right)^{n+1}.$$

Lemma 1 (Bernoulli). For any $\alpha > -1$ and $n \in \mathbb{N}$

$$(1 + \alpha)^n \geq 1 + n\alpha.$$

Proof. (Lemma 1). Let us use the induction by n . For $n = 1$ the lemma is trivial.

Suppose, that it is valid for some $n = k$, that is,

$$(1 + \alpha)^k \geq 1 + k\alpha.$$

To prove it for $n = k + 1$ we can write

$$(1 + \alpha)^{k+1} = (1 + \alpha)^k (1 + \alpha) \geq (1 + k\alpha) (1 + \alpha) = 1 + \alpha + k\alpha + k\alpha^2 \geq 1 + \alpha(k+1).$$

Lemma follows. □

Lemma 2. The sequence $\{y_n\}$ is decreasing and $y_n \geq 2$.

Proof. (Lemma 2). By Bernoulli's lemma

$$y_n = \left(1 + \frac{1}{n}\right)^{n+1} \geq 1 + \frac{n+1}{n} = 1 + 1 + \frac{1}{n} > 2.$$

Also,

$$\begin{aligned} \frac{y_n}{y_{n-1}} &= \frac{\left(1 + \frac{1}{n}\right)^{n+1}}{\left(1 + \frac{1}{n-1}\right)^n} = \frac{\left(1 + \frac{1}{n}\right)^{n+1}}{\left(\frac{n}{n-1}\right)^n} = \left(1 + \frac{1}{n}\right) \left(1 + \frac{1}{n}\right)^n \left(\frac{n-1}{n}\right)^n = \\ &= \left(1 + \frac{1}{n}\right) \left(1 - \frac{1}{n^2}\right)^n. \end{aligned}$$

As

$$\left(1 + \frac{1}{n}\right) = \left(1 + n \cdot \frac{1}{n^2}\right) \leq \left(1 + \frac{1}{n^2}\right)^n$$

by Bernoulli's lemma, we get

$$\frac{y_n}{y_{n-1}} \leq \left(1 + \frac{1}{n^2}\right)^n \left(1 - \frac{1}{n^2}\right)^n = \left(1 - \frac{1}{n^4}\right)^n < 1.$$

Lemma 2 follows. $\square$

So, the sequence $\{y_n\}$ is bounded and decreasing. According to Weierstrass theorem it converges. We can conclude that $\{x_n\}$ is also convergent as a product of two convergent sequences. Theorem follows. $\square$

So, by definition, we put

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e.$$

2. Cantor's theorem.

Consider a sequence of segments

$$[a_1, b_1], [a_2, b_2], \dots, [a_n, b_n], \dots$$

such that

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \dots \supseteq [a_n, b_n] \supseteq \dots, \quad (2)$$

In other words, for any $n \in \mathbb{N}$ ends of segments satisfy the condition

$$a_n \leq a_{n+1} \leq b_{n+1} \leq b_n.$$

Definition 1. If $\lim_{n \rightarrow \infty} (b_n - a_n) = 0$ then the sequence (2) is called *sequence of nested segments*.

Theorem 2 (Cantor's theorem). *For any sequence of nested segments there exists the unique point which belongs to all the segments of the sequence.*

Proof. It is easy to see that left ends of all segments form the nondecreasing sequence

$$a_1 \leq a_2 \leq \dots \leq a_n \leq \dots$$

bounded above by b_1 and below by a_1 . In the same way, right ends of all segments form the increasing sequence

$$b_1 \geq b_2 \geq \dots \geq b_n \geq \dots$$

bounded below by a_1 and above by b_1 . So, both sequences $\{a_n\}$ and $\{b_n\}$ are monotone and bounded. Therefore, they are convergent. Assume,

$$\lim_{n \rightarrow \infty} a_n = a, \quad \lim_{n \rightarrow \infty} b_n = b.$$

By definition 1, $\lim_{n \rightarrow \infty} (b_n - a_n) = 0$. On the other hand,

$$\lim_{n \rightarrow \infty} (b_n - a_n) = \lim_{n \rightarrow \infty} b_n - \lim_{n \rightarrow \infty} a_n = b - a.$$

So, $a = b = C$ and

$$a_n \leq C \leq b_n \tag{3}$$

for $\forall n \in \mathbb{N}$.

Let us now prove the uniqueness. Suppose, that there exists another point $C' \neq C$ which belongs to all the segments (2), that is,

$$a_n \leq C' \leq b_n \tag{4}$$

for $\forall n \in \mathbb{N}$. As both C and C' satisfy conditions (3), (4)

$$|C - C'| \leq b_n - a_n \quad \text{for all } n \in \mathbb{N}$$

Since $\lim_{n \rightarrow \infty} (b_n - a_n) = 0$, we conclude that $C = C'$. Theorem 2 follows. $\square$

Remark. Cantor's theorem is not valid for intervals. Indeed, let us consider the sequence of intervals

$$(0, 1) \supset \left(0, \frac{1}{2}\right) \supset \left(0, \frac{1}{2^2}\right) \supset \dots \supset \left(0, \frac{1}{2^n}\right) \supset \dots$$

and assume that there exists $c \in (0, 1)$ to be common for all intervals $(0, \frac{1}{2^n})$, $n = 1, 2, \dots, n, \dots$. As $c > 0$ we can find $N \in \mathbb{N}$ such that $1/2^N < c$ so that for all $n > N$ intervals $(0, \frac{1}{2^n})$ do not contain c .

3. Bolzano-Weierstrass theorem.

Consider a sequence $\{x_n\}$.

Definition 2. A new sequence $\{y_k\}$ such that $y_k = x_{n_k}$ for

$$n_1 < n_2 < \dots < n_k < \dots$$

is called a *subsequence* of $\{x_n\}$.

First of all, let us formulate the evident

Theorem 3. *If a sequence $\{x_n\}$ is convergent, that is, $\lim_{n \rightarrow \infty} x_n = x$ then its any subsequence $\{y_k\}$ is also convergent and $\lim_{k \rightarrow \infty} y_k = x$.*

Also we can prove the following important result

Theorem 4 (Bolzano-Weierstrass theorem). *Any bounded sequence has a convergent subsequence. In other words, if a sequence $\{x_n\}$ is bounded there exists its convergent subsequence $\{y_k\}$.*

Proof. As $\{x_n\}$ is bounded then there exists such a number M that $|x_n| < M$ for any $n \in \mathbb{N}$. All elements of $\{x_n\}$ belongs to the segment $\Delta_0 = [-M; M]$. Let us divide Δ_0 by two equal parts and let Δ_1 be the part containing infinitely many elements of $\{x_n\}$. We can choose any $x_{n_1} \in \Delta_1$ and put $y_1 = x_{n_1}$.

Next, we can divide Δ_1 by two equal parts again. Let Δ_2 be the part of Δ_1 containing infinitely many elements of $\{x_n\}$. Choose any $x_{n_2} \in \Delta_2$ and put $y_2 = x_{n_2}$ and so on.

So, we get a subsequence $\{y_k\}$ of $\{x_n\}$ and sequence of nested segments

$$\Delta_1 \supseteq \Delta_2 \supseteq \dots$$

The length of Δ_k equals $2M/2^k \rightarrow 0$. So, according to Cantor theorem there exists unique $C \in \Delta_k$ for any $k \in \mathbb{N}$. As $y_k \in \Delta_k$ then $\lim_{k \rightarrow \infty} y_k = C$ and $\{y_k\}$ converges. $\square$